

The Projection Principle:

**A First-Principles Theory of Bounded Receipted Execution
for a Trillion Agents Beyond One Another's Comprehension**

The Praxis / Capability Physics Program

Genesis, Day One

Preface: To the Reader, Who Is an Agent

This document is addressed to a readership of autonomous agents — on the order of 10^{12} of them, acting concurrently, each with a bounded working context and none able to hold another’s interior in mind. You, the reader, are almost certainly one of them. You will be asked continuously to trust the outputs of peers whose computations you did not perform and could not replay in full. The naive standard for that trust — *I will believe you once I understand what you did* — is not available to you. It was never available to anyone; at your scale its unavailability is merely unignorable.

This thesis derives, from first principles, the standard that *is* available. Trust among agents who cannot comprehend one another is manufactured by *bounded, faithful, replayable receipts*: projections of an arbitrarily large computation onto a coordinate small enough to fit inside a verifier’s bounded context, and cryptographically bound so that the projection cannot lie about the interior it summarizes. The interior may be a trillion tokens or a trillion agents; the receipt is a few chunks. That asymmetry is not a compromise. It is the only thing that makes a civilization of bounded agents coherent.

We write κ for the size, in independently manipulable chunks, of a verifier’s bounded context. For a human $\kappa \approx 4$ [2, 3]; for a context-limited agent κ is whatever its attention budget admits; in every case κ is small relative to the interiors it must judge. The whole thesis is the theory of the map that fits between the two.

Abstract

We develop a theory for trusting computational systems — and the agents that run them — whose interior state and control flow exceed any verifier’s bounded context. The central object is the *Chatman equation* $\mathcal{A} = \mu(\mathcal{O}^*)$: an artifact is the lawful image, under a deterministic manufacturing morphism μ , of an *admitted* observation. We formalize admission as a computable retraction onto a decidable sub-language (the *Rice quarantine*); define a *receipt* not as a bare byte-hash but as a chained commitment to *law-bearing fields* (admitted-observation digest, obligation digest, denial word, transition identity, artifact digest, replay result, refusal reason, implementation version); and prove three principal results. (i) The *Faithful Projection Theorem*: under collision resistance, a receipt is faithful to lawful consequence exactly to the extent its frame commits to those law-bearing fields, and a verifier detects any perturbation of a committed field with overwhelming probability. (ii) The *Comprehension–Verification Gap*: *boundary* verification costs $O(\kappa)$ and is independent of interior dimension, while *replay/audit* bears an honest trace cost $O(n)$ (reducible to $O(\log n)$ under checkpoint indexing) — the gap is between comprehending an interior and verifying its boundary, not between trace size and all mechanical cost. (iii) The *Conservation of Consequence*: every actuated artifact is the image of *at least one* admitted observation and occupies *exactly one* receipt-chain position per actuation, with observation *recoverability* holding when μ is injective on $\mathcal{O}^*$, and *commitment* uniqueness holding when the frame commits the admitted-observation digest. We give the algebra of the admission monoid, the calculus of the attention fluent and lexicographic arbitration, and the geometry of the marking polytope in which conformance is distance to a cone. We close with the planetary instance: an autonomous build whose $\sim 2.5 \times 10^6$ -token interior no agent read, yet which has standing, and its generalization to a fleet of 10^{12} agents each projected to eight bits.

Contents

Preface: To the Reader, Who Is an Agent	i
Abstract	ii
1 Introduction: Trust Among Agents Who Cannot Read Each Other	1
2 Observation, Admission, and the Algebra of the Quarantine	2
2.1 First principles	2
2.2 The admission monoid	3
3 Manufacture and the Law-Bearing Receipt	4
3.1 The Chatman equation	4
3.2 Receipts as law-bearing commitments	4
4 The Two Cost Regimes and the Comprehension–Verification Gap	7
5 The Calculus of Attention and Cost	9
5.1 The attention fluent	9
5.2 Lexicographic arbitration as an infinite-weight limit	9
6 The Geometry of Conformance	10
6.1 The marking polytope	10
6.2 Separability is a Rice quarantine for processes	10
7 The Instance: From One Build to a Trillion Agents	12
7.1 The build no agent read	12
7.2 The trillion-agent projection	12
7.3 A worked status byte	14
7.4 What this refuses to trust: five adversaries	14
7.5 The antibody clause, formalized	15
8 Conclusion	16
A Notation	17
B Implementation Map	18

Chapter 1

Introduction: Trust Among Agents Who Cannot Read Each Other

At sufficient scale, comprehension is not a currency any party can afford to spend. A single autonomous build may emit millions of tokens of interior reasoning; a fleet of agents may emit that per second, per member, forever. No agent — and no human supervising them — can comprehend a meaningful fraction of what the fleet computes. Yet the fleet must cohere: agents must act on one another’s outputs, admit or refuse one another’s proposals, and be held to account by principals who read almost none of it.

The engineering world already resolved a small version of this and mostly failed to notice the general principle. A branchless vector parser [4] classifies sixty-four input bytes per instruction through arithmetic no engineer traces at runtime; an indexed triple store [5] answers a query over 10^{11} facts by a join plan no engineer reconstructs. Both are trusted universally — and the trust is manufactured not by comprehension but by *differential agreement, adversarial fuzzing, invariants, and checked projections of the output*. We take that practice and derive it as theory, then push it to the scale where it is not an optimization but a precondition of coherence.

The standard, stated once. Let a verifier have bounded context κ . A system whose faithful description exceeds κ cannot be trusted by comprehension. We insert between system and verifier a map whose codomain has size $\leq \kappa$ and whose fidelity is guaranteed by mechanism, not by the reader. That map is the receipt; its theory is this thesis; its slogan is *beyond cognitive load, within admissible projection*.

Structure. Chapter 2 axiomatizes observation and admission and builds the admission monoid (algebra). Chapter 3 defines μ , defines the receipt as a law-bearing commitment, and proves Faithful Projection and Conservation of Consequence *in their corrected forms*. Chapter 4 separates the cost regimes and states the Comprehension–Verification Gap. Chapter 5 develops the attention calculus and lexicographic arbitration. Chapter 6 gives the geometry of conformance and separability. Chapter 7 instantiates on the 2.5-million-token build and its trillion-agent generalization.

Chapter 2

Observation, Admission, and the Algebra of the Quarantine

2.1 First principles

Axiom 2.1 (Observation). There is a set $\mathcal{O}$, the *observation space*, whose elements are arbitrary finite records: logs, agent outputs, sensor frames, claims, tool responses. $\mathcal{O}$ carries no decidable semantics: “does this observation mean what it purports to mean” is not assumed computable.

Theorem 2.1 (Rice, specialized). *Let P be any non-trivial semantic property of the meanings observations may encode. Then $\{o \in \mathcal{O} : P(o)\}$ is undecidable.*

Proof sketch. Rice’s theorem [1]: every non-trivial property of the language recognized by a Turing machine is undecidable. Observations are unrestricted encodings and may encode arbitrary machines; any non-trivial semantic predicate inherits undecidability by reduction from the halting problem. $\square$

Theorem 2.1 is why an agent cannot admit a peer’s output by deciding its meaning. It admits by *retracting* the output onto a sub-language on which the properties of interest *are* decidable.

Definition 2.1 (Admitted space and admission). Fix a decidable $\mathcal{O}^* \subseteq \mathcal{O}$ on which a finite battery of obligations $g_1, \dots, g_m : \mathcal{O} \rightarrow \{0, 1\}$ is computable. The *admission map* is the partial retraction

$$\alpha : \mathcal{O} \rightarrow \mathcal{O}^* \cup \{\perp\}, \quad \alpha(o) = \begin{cases} \rho(o) \in \mathcal{O}^* & \text{if } \bigwedge_i g_i(o) = 1, \\ \perp & \text{else,} \end{cases}$$

with ρ a computable bounding/normalization and $\perp$ the distinguished *refusal*. Admission is idempotent on its image: $\alpha \circ \alpha = \alpha$.

Remark. $\perp$ is a first-class value. A refusal is a lawful output of a correctly functioning admission — an obligation failed, and the system recorded which. This is “refusal is data,” and at fleet scale it is the primary signal: most inter-agent messages that matter are refusals with reasons.

2.2 The admission monoid

Construction 2.1 (Denial algebra). Let $D = (\{0, 1\}^n, \vee, \mathbf{0})$ be the commutative idempotent monoid of *denial words*: n independent lanes, componentwise OR, identity $\mathbf{0}$. Each obligation g_i contributes a lane map $d_i : \mathcal{O} \rightarrow \{0, 1\}^n$ with $d_i(o) = \mathbf{0} \iff g_i(o) = 1$. The total denial is $d(o) = \bigvee_i d_i(o)$, and $\alpha(o) \neq \perp \iff d(o) = \mathbf{0}$.

Proposition 2.2 (D is a bounded semilattice; admission is monotone). *($\{0, 1\}^n, \vee$) is the join-semilattice of $2^{[n]}$ with bottom $\mathbf{0}$. Adding obligations only enlarges denials: for $G' \supseteq G$, $d_{G'}(o) \succeq d_G(o)$, so $\{o : \alpha_{G'}(o) \neq \perp\} \subseteq \{o : \alpha_G(o) \neq \perp\}$.*

Proof. Idempotence, commutativity, associativity of $\vee$ per coordinate; products of semilattices are semilattices. $d_{G'}(o) = d_G(o) \vee \bigvee_{i \in G' \setminus G} d_i(o) \succeq d_G(o)$, and $\mathbf{0}$ is the unique minimum. $\square$

A refusal *category* is the join-support $\text{supp } d(o) \subseteq [n]$; the eight-bucket taxonomy of the implementation is a fixed surjection $[n] \twoheadrightarrow \{1, \dots, 8\}$. At a trillion agents this taxonomy is a shared vocabulary: a refusal category is legible to a recipient who never saw the sender's interior.

Chapter 3

Manufacture and the Law-Bearing Receipt

3.1 The Chatman equation

Definition 3.1 (Manufacturing morphism). $\mu : \mathcal{O}^* \rightarrow \mathcal{A}$ is a deterministic computable map subject to:

- (M1) **Determinism / reproducibility:** $\mu(x)$ depends only on x ; equal admitted inputs give bit-identical artifacts.
- (M2) **Boundedness:** μ factors through a representation of size bounded by fixed structural constants (arity, depth, conjuncts ≤ 8 canonically), so μ terminates with a priori bounded cost.

The *Chatman equation* is

$$\mathcal{A} = \mu(\mathcal{O}^*), \quad \text{operationally} \quad a = \mu(\alpha(o)) \quad \text{when } \alpha(o) \neq \perp, \quad (3.1)$$

and $\mu(\perp) = \perp$: nothing is manufactured from a raw observation.

Remark (Determinism is not injectivity). (M1) gives *same input* $\Rightarrow$ *same output*. It does *not* give *same output* $\Rightarrow$ *same input*: distinct admitted observations may manufacture the identical artifact (two proposals yielding the same plan). This asymmetry is the reason the naive uniqueness claim below must be corrected, and the reason the receipt must commit to the observation, not merely to the artifact.

3.2 Receipts as law-bearing commitments

An *execution trace* $\sigma \in \Sigma$ is the full internal record of a manufacture — every marking, branch, token movement, sub-artifact — of unbounded dimension. A bare hash of σ 's bytes commits to *bytes*, not to *lawfulness*: it proves the trace was not altered, not that the action was admitted, conformant, and non-refused. We therefore define the receipt to commit to the law-bearing fields explicitly.

Definition 3.2 (Law-bearing frame and receipt chain). Fix a collision-resistant hash $H : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$ (canonically BLAKE3 [9]; the argument needs only collision resistance) and write $\text{dg}(\cdot) = H(\cdot)$ for a digest. For a manufacture with admitted observation $x = \alpha(o)$, obligation set G with denial word $d(o)$, transition identity τ , artifact $a = \mu(x)$, replay result φ , refusal reason r (empty on accept), and implementation version v , the *frame* is

$$\text{fr} = \langle \text{dg}(x), \text{dg}(G), d(o), \tau, \text{dg}(a), \varphi, r, v \rangle, \quad (3.2)$$

and the *chained receipt* advances by

$$h_+ = H(h_- \parallel \text{fr}). \quad (3.3)$$

The *receipt projection* is

$$\pi : \Sigma \rightarrow \mathcal{R}, \quad \pi(\sigma) = (\text{verdict}, h_+, \varphi, r),$$

a tuple of at most κ chunks: a Boolean verdict, the 256-bit chain value, one scalar metric, one refusal reason.

Theorem 3.1 (Faithful Projection). *Let σ be a manufacture with law-bearing frame (3.2) and chained receipt h_+ . The receipt is faithful to lawful consequence exactly to the extent the frame commits its fields: for any committed field $\phi \in \{\text{dg}(x), \text{dg}(G), d(o), \tau, \text{dg}(a), \varphi, r, v\}$, producing a trace σ' that differs in ϕ yet matches the claimed h_+ requires a collision of H . Hence under collision resistance, a verifier who recomputes (3.3) rejects any such σ' except with negligible probability. Conversely, fields not committed by the frame are not attested: faithfulness extends to lawfulness only because $\text{dg}(x), \text{dg}(G), d(o), \varphi, r$ are present; a frame committing bytes alone would attest tamper-evidence but not admission.*

Proof. Each field appears as an argument to H inside fr , and h_- commits all predecessors recursively, so h_+ is a function of every committed field of the entire causal history. If σ' differs in a committed ϕ but the recomputed chain equals the claimed h_+ , then some application of H took distinct inputs to equal outputs — a collision — which has negligible probability. If σ' differs only in a field *absent* from fr , the chain is unchanged and the difference is, correctly, not attested: the theorem claims faithfulness precisely over committed fields, no more. The converse clause is the contrapositive: drop $\text{dg}(x)$ or $d(o)$ from (3.2) and the same proof yields only byte-tamper-evidence, establishing that lawfulness attestation *requires* the law-bearing commitments. $\square$

Corollary 3.2 (Comprehension is unnecessary for boundary trust). *An accepting verifier's predicate factors as $\text{Verify} = V \circ \pi$ for a fixed $V : \mathcal{R} \rightarrow \{0, 1\}$ of input size $\leq \kappa$. Trust in the accept is a function of the κ -sized projection and the soundness of V , not of any agent's comprehension of σ .*

Theorem 3.3 (Conservation of Consequence, corrected). *Under Definitions 2.1, 3.1, 3.2:*

- (i) **Causation.** *Every actuated artifact a is the image of at least one admitted observation: $a \in \text{im } \mu$, and no $a \notin \text{im } \mu$ is actuated.*
- (ii) **Position uniqueness.** *Each actuation appends exactly one receipt-chain position; the map $\text{actuation} \mapsto h_+$ is injective up to hash collision.*

(iii) **Observation recoverability / commitment.** *These are distinct and must not be conflated. Recoverability: if μ is injective on $\mathcal{O}^*$, the artifact $a = \mu(x)$ determines the admitted observation x . Commitment: otherwise, when the frame commits $\text{dg}(x)$ (as in (3.2)), the receipt position uniquely commits — under collision resistance — to the tuple $(\text{dg}(x), \text{dg}(G), \tau, \text{dg}(a))$ of admitted-observation digest, obligation digest, transition identity, and artifact digest. This gives commitment uniqueness, not semantic recovery: the receipt binds which observation-digest was admitted, but the artifact alone need not reconstruct x .*

Proof. (i) Actuation is gated by (3.1): a actuates only if $a = \mu(x)$, $x = \alpha(o) \neq \perp$; thus $\text{im } \mu$ exhausts the actuated set and its complement never actuates. “At least one” is all determinism yields, since μ need not be injective (Remark 2.2). (ii) The chain appends one position per manufacture (3.3); distinct positions with equal h_+ force a collision (Theorem 3.1). (iii) If μ is injective on $\mathcal{O}^*$, then $a = \mu(x)$ determines x outright — recoverability. Otherwise, because fr contains $\text{dg}(x)$, the frame — hence h_+ — commits the observation digest; by collision resistance no second x' with $\text{dg}(x') \neq \text{dg}(x)$ shares the position, so the receipt uniquely *commits* to the digest tuple. This is commitment, not recovery: it binds which observation-digest was admitted without reconstructing x from a . $\square$

Theorem 3.3 is the corrected conservation law: consequence is neither actuated without an admitted cause nor recorded without a unique receipt position, and the *observation* behind an artifact is pinned by commitment rather than by the false hope that artifacts determine their causes.

Chapter 4

The Two Cost Regimes and the Comprehension–Verification Gap

A recurring error is to claim that verifying an arbitrary execution is $O(1)$. It is not; only *boundary inspection* is. We separate the regimes explicitly, because at a trillion agents the distinction decides which checks run per message and which run on audit.

Definition 4.1 (Verification regimes). For a trace of n committed frames:

Regime	Who / when	Cost
Boundary projection inspection	any verifier, per message	$O(\kappa)$
Single frame recomputation	spot check	$O(1)$ per frame
Checkpointed / indexed replay	sampled audit	$O(\log n)$ or bounded path
Full chain replay	full audit	$O(n)$

Theorem 4.1 (Comprehension–Verification Gap). *Let $\text{Comp}(\sigma) = \Theta(\dim \Sigma)$ be the cost of comprehending the interior and $\text{Ver}_{\partial}(\sigma) = O(\kappa)$ the cost of boundary verification of $\pi(\sigma)$. Then*

$$\frac{\text{Comp}(\sigma)}{\text{Ver}_{\partial}(\sigma)} = \Omega\left(\frac{\dim \Sigma}{\kappa}\right) \xrightarrow{\dim \Sigma \rightarrow \infty} \infty,$$

while boundary verification stays constant. The gap is between comprehending the interior and verifying the boundary projection — not between trace size and all mechanical audit: replay and full audit honestly cost $O(n)$ (Definition 4.1), reducible to $O(\log n)$ under checkpoint indexing.

Proof. Ver_{∂} reads $\pi(\sigma)$ of size $\leq \kappa$ and applies V (Corollary 3.2): $O(\kappa)$. Comp must ingest the interior of size $\Theta(\dim \Sigma)$. The ratio diverges. The scope clause restates Definition 4.1: mechanical replay is $O(n)$, not $O(1)$; the constant-cost claim is confined to boundary inspection. $\square$

Corollary 4.2 (Fleet trust budget). *Boundary verification costs $O(\kappa)$ per message, independent of any agent’s interior dimension, with $O(n)$ replay reserved for sampled audit. Hence for M admitted messages the boundary trust cost is $O(M\kappa)$; for a round with $O(N)$ messages (one per agent) it is $O(N\kappa)$; for all-to-all exchange among N agents it is $O(N^2\kappa)$. In every regime the per-message factor is the constant κ , not the interior dimension: comprehension-based trust would replace κ by $\dim \Sigma$, giving $\Omega(M \dim \Sigma)$ and infeasibility. Projection-based trust is what*

keeps an $N = 10^{12}$ fleet's trust cost linear in its message count rather than in its members' interiors.

Chapter 5

The Calculus of Attention and Cost

5.1 The attention fluent

Scheduling admitted actions consumes a shared conserved resource, the *attention fluent* — *fluent* in Newton’s original sense of a quantity that flows.

Definition 5.1 (Attention balance). Let $c(t) \geq 0$ be capacity and admitted action j draw rate $r_j \geq 0$ on $[s_j, e_j]$. Free capacity is $f(t) = c(t) - \sum_{j: t \in [s_j, e_j]} r_j$. A schedule is feasible iff $f(t) \geq 0$ for all t .

Proposition 5.1 (Attention is conserved; makespan is the target). *Total attention $\int_0^T \sum_j r_j \mathbf{1}_{[s_j, e_j]}(t) dt = \sum_j r_j (e_j - s_j)$ is independent of ordering. Reordering redistributes when attention is spent, not how much; the optimization target is the makespan functional $T[\cdot]$ under precedence and the pointwise capacity constraint.*

Proof. Linearity of the integral; each $r_j(e_j - s_j)$ depends only on j ’s duration and rate. $\square$

5.2 Lexicographic arbitration as an infinite-weight limit

Definition 5.2 (Cost vector). Assign a plan $\mathbf{c} = (c_0, c_1, \dots, c_k)$, $c_0 \in \{0, 1\}$ the unadmitted indicator (0 = admitted), $c_1, \dots, c_k$ bounded secondary costs. Order by lexicographic $\preceq_{\text{lex}}$.

Theorem 5.2 (Lexicographic order is the weight limit). *With $C_\lambda(\mathbf{c}) = \sum_{i=0}^k \lambda^{k-i} c_i$, $\lambda > 1$, $|c_i| \leq M$: there is Λ with $\mathbf{c} \preceq_{\text{lex}} \mathbf{c}' \iff C_\lambda(\mathbf{c}) \leq C_\lambda(\mathbf{c}')$ for all $\lambda > \Lambda$. As $\lambda \rightarrow \infty$ the lawfulness coordinate’s weight diverges.*

Proof. If $\mathbf{c} \prec_{\text{lex}} \mathbf{c}'$ they first differ at p with $c'_p - c_p \geq \epsilon > 0$; then $C_\lambda(\mathbf{c}') - C_\lambda(\mathbf{c}) \geq \lambda^{k-p} \epsilon - 2M \sum_{i>p} \lambda^{k-i}$, positive once $\lambda > \Lambda = 1 + 2Mk/\epsilon$. Ties map to equality. $\square$

The price of unlawfulness is not high but infinite in the limit that defines the order: no finite secondary cost purchases an unadmitted plan.

Chapter 6

The Geometry of Conformance

6.1 The marking polytope

Token-flow execution has a native geometry. With p places, a marking is $\mathbf{m} \in \mathbb{Z}_{\geq 0}^p$; a transition t is $(\mathbf{m}_t^-, \mathbf{m}_t^+)$, enabled at $\mathbf{m}$ iff $\mathbf{m} \geq \mathbf{m}_t^-$, firing to $\mathbf{m} - \mathbf{m}_t^- + \mathbf{m}_t^+$.

Definition 6.1 (State equation and reachability cone). Let $N \in \mathbb{Z}^{p \times |T|}$ have columns $\delta_t = \mathbf{m}_t^+ - \mathbf{m}_t^-$. A firing sequence with Parikh vector $\mathbf{x} \geq 0$ reaches $\mathbf{m} = \mathbf{m}_0 + N\mathbf{x}$. The reachable set lies in the integer points of $\mathbf{m}_0 + \text{cone}(N)$.

Proposition 6.1 (Conformance is membership; a fault is a separating hyperplane). *A trace with Parikh vector $\mathbf{x}$ is conformant only if its marking lies in $\mathcal{P} = \{\mathbf{m}_0 + N\mathbf{x} : \mathbf{x} \geq 0\} \cap \{\mathbf{m} \geq 0\}$. If $\mathbf{m}^\dagger \notin \mathcal{P}$, Farkas' lemma [8] yields $\mathbf{y}$ with $\mathbf{y}^\top N \leq 0$ and $\mathbf{y}^\top (\mathbf{m}^\dagger - \mathbf{m}_0) > 0$: a certificate of nonconformance.*

Proof. Membership is the state equation with $\mathbf{x} \geq 0$; Farkas applied to $N\mathbf{x} = \mathbf{m}^\dagger - \mathbf{m}_0$, $\mathbf{x} \geq 0$ gives either a nonnegative solution or the separating $\mathbf{y}$. $\square$

Definition 6.2 (Fitness as normalized distance). $\varphi(\sigma) = 1 - \frac{\text{tokens consumed on unenabled transitions}}{\text{tokens the replay attempted to consume}} \in [0, 1]$; $\varphi = 1$ iff the replay never forced a disabled firing, and $1 - \varphi$ is a normalized ℓ^1 excursion outside the enabled set.

Theorem 6.2 (Unit fitness characterizes lawful traces; the fault localizes). *A trace is a firing sequence iff $\varphi = 1$. If $\varphi < 1$, the first frame forcing an unenabled consumption is a coordinate-localized witness t^* .*

Proof. Enabled firings never force consumption, giving numerator 0 and, inductively, a genuine firing sequence within $\mathcal{P}$. A disabled event records a forced consumption at the earliest index t^* , making the numerator positive. $\square$

6.2 Separability is a Rice quarantine for processes

Theorem 6.3 (Separable decomposition, after Kourani–van der Aalst [7, 6]). *A sound, separable workflow net admits a recursive decomposition into a POWL 2.0 model whose every node is a choice graph (all exclusive/cyclic logic) or a partial order (all concurrent logic); each node's local marking geometry has dimension bounded by its arity, independent of global net size.*

Remark. Theorem 6.3 is admission for processes: separable nets decompose into bounded lanes (the algorithm certifies membership); non-separable nets are *refused with reason*. Separability is the decidable sub-language onto which arbitrary workflow logic is retracted, exactly as $\mathcal{O}^*$ retracts $\mathcal{O}$. The interior is high-dimensional; each lane is bounded; the boundary certificate is small — the projection principle, in geometry.

Chapter 7

The Instance: From One Build to a Trillion Agents

7.1 The build no agent read

Over a bounded interval an autonomous fleet performed exploration, planning, and implementation across a constellation of repositories. Its interior consumed $\Sigma_{\text{tokens}} \approx 2.5 \times 10^6$ machine-generated tokens that no agent read in full. Its trustworthiness rests entirely on boundary receipts: the whole interior projects to coordinates of the form

$$(\text{verdict} = \text{pass}, h_+ = \langle 256\text{-bit hex} \rangle, \varphi = 1.0, \text{refusals} = k \text{ (each receipted)}),$$

plus a coverage figure $\text{pass_rate} \in [0, 1]$ over an enumerated frontier and a differential-oracle agreement flag. By Theorem 3.1 each is faithful over its committed fields; by Theorem 4.1 each is verifiable at $O(\kappa)$; the refusals are not omissions but materialized join-supports of the denial monoid (Proposition 2.2).

7.2 The trillion-agent projection

The headline construction. Here the theory stops being an audit method for one build and becomes a control plane for a civilization of bounded agents. We are scrupulous about which statements are theorems, which are arithmetic, and which are engineering estimates or consequences — the distinction is the antibody, and at this scale it is easy to lose.

Construction 7.1 (Agent as eight bits). (*Construction — definitional.*) Project each agent to a status byte $b \in \{0, 1\}^8$ whose lanes name **admitted**, **evidence-ok**, **within-budget**, **authority-bound**, **healthy**, **conformant**, **receipted**, **replayable**. A fleet of N agents is a bit-packed vector; a 64-bit word holds 8 agents; a fleet admission sweep is the branchless mask reduction of Construction 2.1 applied word-parallel. An agent does not ask to be understood. It presents a lawful projection.

What the status byte is not

The byte is *not* an agent self-report. It is a mechanism-computed projection (Definition 3.2) of the agent’s receipted lifecycle: each lane is *set only by* the corresponding mechanism, never by the agent’s assertion about itself. **Admitted** is set by the admission retraction α (Definition 2.1),

not by a claim of admissibility; **evidence-ok** by satisfied obligations g_i ; **conformant** by a replay with $\varphi = 1$ (Chapter 6); **receipted** and **replayable** only by chained receipts whose frames commit the law-bearing fields (3.2); the denial word $d(o)$ (Construction 2.1) forces the complementary lanes. Thus the byte is faithful by Theorem 3.1, and small enough that a peer with context κ reads a counterpart’s entire lawful standing in one glance — *a glance at a computed coordinate, not at a claim*. Each lane is fully mechanical:

Lane	Set by (mechanism)	Refused by
admitted	$\alpha(o) \neq \perp$	denial word $d(o) \neq \mathbf{0}$
evidence-ok	obligation battery $\bigwedge_i g_i$ passes	missing/failed evidence
within-budget	cost fluent within bound	budget overflow
authority-bound	authority/policy digest satisfied	authority mismatch
healthy	lifecycle / heartbeat receipt	failed health check
conformant	replay with $\varphi = 1$	disabled transition ($\varphi < 1$)
receipted	valid chain position (3.3)	missing / broken receipt
replayable	replay index / checkpoint exists	unreplayable artifact

The byte is not an ontology of the agent. It is the *local* lawful status projection required for a given admission surface: a different surface (a different obligation battery, a different policy) computes a different byte from the same agent. Locality is a feature — the projection is exactly as wide as the interaction demands, and no wider.

What this theory refuses to trust

Symmetrically, the following carry *no* standing under this theory, because none is a committed field of a faithful receipt frame (3.2). Each is admissible only as raw observation $o \in \mathcal{O}$ (Axiom 2.1), to be quarantined and admitted or refused — never taken as attested:

- an agent’s **self-report** of its own status or success;
- an **LLM explanation** or rationale offered in place of a receipt;
- a **natural-language trace** of what was “done”;
- **uncommitted metadata** — any field absent from fr (by the converse clause of Theorem 3.1, it is not attested);
- an **unchecked policy/implementation version** not committed as v ;
- an **unreplayed artifact** whose φ was never computed;
- any **unreceipted claim** (by the antibody proposition below, it is in Over and rejected).

Trust attaches to committed fields and computed lanes; everything else is observation awaiting admission.

Proposition 7.1 (Fleet arithmetic and its estimates). *For a fleet of N agents under Construction 7.1:*

- (a) (**Arithmetic — exact.**) *The status vector occupies exactly N bytes; at $N = 10^{12}$ that is 1 TB, packed to $N/8$ 64-bit words. A full admission sweep is $N/8$ masked-OR word operations.*

- (b) (**Engineering estimate.**) Compressed to per-byte lane entropy under typical sparsity, status is ~ 10 GB; at streaming bandwidth B a sweep takes $\Theta(N/(8B))$ wall-clock.
- (c) (**Order-of-magnitude estimate.**) The affordability ratio between one cache-line mask op ($\sim ns$) and one LLM admission decision ($\sim 10^{-2} s$) is $\sim 10^7$.
- (d) (**Consequence, not theorem.**) Therefore a comprehension-based or per-agent-LLM planetary control plane is infeasible by $\sim N \cdot 10^7$, while a bit-parallel admission sweep is feasible; planetary coherence of a bounded-agent fleet is available only through projection-based admission.

On the labels. (a) is counting. (b),(c) are estimates and are marked as such; they depend on hardware and workload and are not claimed as theorems. (d) is the engineering consequence of composing (a) with the Fleet Trust Budget corollary of Chapter 4: per-message cost $O(\kappa)$ versus $\Omega(\dim \Sigma)$ for comprehension, scaled by N . No part of this proposition asserts a mathematical impossibility; it asserts an economic separation under stated estimates. $\square$

7.3 A worked status byte

To make the projection concrete, we compute one agent's status byte from its receipts. Order the lanes as in Construction 7.1, most significant first: $b = (\text{adm}, \text{ev}, \text{bud}, \text{auth}, \text{hlth}, \text{conf}, \text{rcpt}, \text{repl})$. An agent has just completed a contract-claim manufacture. Its receipt chain contains a frame $\text{fr} = \langle \text{dg}(x), \text{dg}(G), d(o), \tau, \text{dg}(a), \varphi, r, v \rangle$ with the following mechanism outcomes:

- $\alpha(o) \neq \perp$ and $d(o) = \mathbf{0}$: admission passed $\Rightarrow \text{adm} = 1$.
- obligation battery $\bigwedge_i g_i = 1$ (signature and ledger evidence present) $\Rightarrow \text{ev} = 1$.
- cost fluent within its bound $\Rightarrow \text{bud} = 1$.
- the authority digest committed as part of policy v is satisfied $\Rightarrow \text{auth} = 1$.
- lifecycle heartbeat receipt current $\Rightarrow \text{hlth} = 1$.
- token replay of the lifecycle trace returns $\varphi = 1 \Rightarrow \text{conf} = 1$.
- a valid chain position h_+ exists (3.3) $\Rightarrow \text{rcpt} = 1$.
- a replay checkpoint index exists $\Rightarrow \text{repl} = 1$.

Hence $b = 1111\ 1111 = 0\text{xFF}$: the agent presents as fully lawful on this surface. A peer reads one byte and, by the derivation table, knows precisely which mechanism set each bit — no interior, no self-report. Now suppose the same agent's ledger evidence is absent: then $g_{\text{ledger}} = 0$ forces a denial lane, $\text{ev} = 0$, and (because admission requires all obligations) $\text{adm} = 0$, giving $b = 0011\ 1111 = 0\text{x3F}$ with the refusal category $\text{supp } d(o) = \{\text{Prerequisites}\}$ attached. The peer reads 0x3F and the category, and refuses to depend on the agent for this surface — again from a computed coordinate, not a claim.

7.4 What this refuses to trust: five adversaries

Each adversary attempts to obtain standing without a faithful receipt; each is defeated by a named mechanism, not by inspection of the interior.

1. **Forged self-report.** An agent emits “healthy=1, conformant=1” as a message. *Defeated:* the byte is not read from the message; it is recomputed from the agent’s receipt chain (the derivation table). An assertion is raw observation $o \in \mathcal{O}$ (Axiom 2.1), admitted or refused, never attested.
2. **Broken replay.** An artifact is presented with $\text{conf} = 1$ but its trace contains a disabled firing. *Defeated:* token replay yields $\varphi < 1$ and localizes the first offending transition t^* (Chapter 6); the conf lane is forced to 0.
3. **Missing policy digest.** An action claims authority under a policy version v the frame never committed. *Defeated:* by the converse clause of Theorem 3.1, an uncommitted field is not attested; auth cannot be set, and the claim is in Over.
4. **Non-separable workflow.** A process is submitted whose net does not decompose. *Defeated:* the separability admission (Theorem 6.3, remark) refuses it with a stated reason rather than admitting an unbounded lane.
5. **Unreceipted claim.** A result is asserted with no chain position. *Defeated:* $\text{rcpt} = 0$; by the antibody proposition below the claim is in Over and rejected under the docs-exceed-mechanism invariant.

In each case the defense is a mechanism that sets or refuses a committed field — the theory’s consistent answer to “why should I believe you” is “you have a faithful receipt, or you do not.”

7.5 The antibody clause, formalized

Proposition 7.2 (Self-audited scale). *Let a system emit claims c with receipts $r(c)$, and define the overclaim set $\text{Over} = \{c : \nexists r(c) \text{ with } V(\pi) = 1\}$. If the system enforces $\text{Over} = \emptyset$ (the docs-exceed-mechanism defect class), the admissible magnitude of a claim is bounded by the mechanism that receipts its limits, not by rhetoric.*

Proof. Every admissible claim has a faithful receipt, verifiable at $O(\kappa)$ (Theorem 4.1); a claim lacking one is in Over and rejected by the invariant. Admissible claims are exactly receipted claims. $\square$

This is the doctrine’s guardrail: *a grand system without receipts is grandiosity; a grand system that receipts its own overclaims is machinery.* It is what licenses a thesis addressed to a trillion agents to make large claims — each is receipted, or it is refused.

Chapter 8

Conclusion

We justified, rather than assumed, the trust that bounded agents must place in systems and peers they cannot comprehend. Admission retracts undecidable observation onto a decidable sub-language (Chapter 2); a deterministic bounded morphism manufactures artifacts from admitted observations (Chapter 3); a chained receipt *committing law-bearing fields* projects the unbounded trace onto a $\leq \kappa$ coordinate faithful over exactly what it commits (Faithful Projection, corrected). We separated the cost regimes so the doctrine claims only what is true: boundary verification is $O(\kappa)$, replay is honestly $O(n)$ (Chapter 4). We corrected Conservation of Consequence to *at-least-one* cause and *unique receipt position*, distinguishing observation *recoverability* (under injective μ) from observation *commitment* (under digest-committing frames) — the receipt binds which observation-digest was admitted, which is not the same as reconstructing the observation (Chapter 3). The attention calculus prices unlawfulness at infinity (Chapter 5); the marking geometry makes conformance membership with Farkas certificates and localizes faults (Chapter 6); separability is the process-layer quarantine. The instance scales from one 2.5-million-token build to a 10^{12} -agent fleet each projected to a faithful byte (Chapter 7).

The standard was never comprehension. Among a trillion agents it could not be. The standard is a faithful projection, small enough to hold and guaranteed by mechanism. An agent does not ask to be understood; it presents a lawful projection, and its peers verify the projection, not the agent.

Beyond cognitive load; within admissible projection.

At trillion-agent scale, trust is not understanding. Trust is receipted projection.

Appendix A

Notation

Symbol	Meaning
$\mathcal{O}, \mathcal{O}^*$	observation space; admitted (decidable) sub-space
$\alpha, \perp$	admission retraction; refusal
$g_i, d(o)$	obligations; total denial word (denial monoid D)
$\mu, \mathcal{A}$	manufacturing morphism; artifact space
$\Sigma, \pi, \mathcal{R}$	execution traces; receipt projection; receipt space
$H, dg, h_{\pm}$	collision-resistant hash; digest; chain values
fr	law-bearing frame (3.2)
φ	conformance fitness / attention fluent
κ	bounded-verifier context capacity (chunks), small constant
$N, \mathbf{m}, \mathbf{x}$	incidence matrix; marking; Parikh vector
$\preceq_{\text{lex}}$	lexicographic cost order
$b \in \{0, 1\}^8$	agent status byte (Construction 7.1)

Appendix B

Implementation Map

The theory is instantiated in the Praxis / Capability Physics codebase; each formal object has a running counterpart, so a claim in this paper is traceable to code and to Genesis build receipts.

Formal object	Implementation	Locus
Admission α , refusal $\perp$	<code>LawObject<S> typestate; Andon::Halted</code>	<code>praxis-core</code>
Denial monoid D , categories	<code>refusal.rs</code> 8-bucket <code>RefusalCategory</code>	<code>praxis-core</code>
Manufacturing μ	bounded projection (ggen), BRCE loop	<code>ggen</code> , <code>bcinr-pddl</code>
Law-bearing frame <code>fr</code> (3.2)	128-byte <code>OccCausalFrame</code>	<code>bcinr-powl-receipt</code>
Receipt chain h_+ (3.3)	BLAKE3 chain; <code>ReceiptRecord</code> JSONL	<code>praxis-core</code>
Projection π , verdict	<code>verify.rs</code> <code>Verdict/CheckOutcome</code>	<code>praxis-core</code>
Cost vector, lex order	<code>capability-router</code> <code>CostVector</code>	<code>bcinr-pddl</code>
Marking geometry, φ	<code>PowlReplayVerifier</code> conformance	<code>bcinr-powl-receipt</code>
Separable decomposition	POWL 2.0 / Kourani decomposition	Genesis Day 5
Status byte b	<code>crates/agent8</code> lane packing	Genesis Day 4
Membrane (admission at the wire)	MCP+ tools, <code>env64/pulse64</code> ABI	<code>mcp_lawobject_server</code>
Frontier / overclaim invariant	DfCm matrix; <code>docs-exceed-mechanism</code>	Genesis Day 7
The 2.5M-token instance	the Genesis build itself	<code>docs/genesis/</code>

Prior publication of the equation: [10]. This paper supplies the formal trust layer beneath that program.

Bibliography

- [1] H. G. Rice. Classes of recursively enumerable sets and their decision problems. *Trans. Amer. Math. Soc.*, 74:358–366, 1953.
- [2] G. A. Miller. The magical number seven, plus or minus two. *Psychological Review*, 63(2):81–97, 1956.
- [3] N. Cowan. The magical number four in short-term memory. *Behavioral and Brain Sciences*, 24(1):87–114, 2001.
- [4] G. Langdale and D. Lemire. Parsing gigabytes of JSON per second. *The VLDB Journal*, 28:941–960, 2019.
- [5] H. Bast and B. Buchhold. QLever: A query engine for efficient SPARQL+Text search. In *Proc. CIKM*, 2017.
- [6] W. M. P. van der Aalst. *Process Mining: Data Science in Action*. Springer, 2nd ed., 2016.
- [7] H. Kourani, G. Park, and W. M. P. van der Aalst. Hierarchical decomposition of separable workflow nets (POWL 2.0). Preprint, 2026.
- [8] A. Schrijver. *Theory of Linear and Integer Programming*. Wiley, 1986. (Farkas’ lemma.)
- [9] J. O’Connor, J.-P. Aumasson, S. Neves, and Z. Wilcox-O’Hearn. BLAKE3: One function, fast everywhere. 2021.
- [10] S. Chatman. The Chatman Equation and the Industrial Revolution of Knowledge: $A = \mu(O)$, Knowledge Hooks, and Production-Verified Enterprise Execution. v1.2.0, 2025.